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SmartDock

Dock Appointment Scheduling & Yard Management
for Warehouses and Distribution Centres

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Academic year 2026–2027

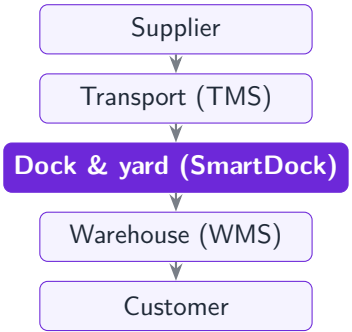
Supply Chain Management – Final Project

Outline

1. Introduction: IT in supply chains
2. Problem statement
3. Objectives
4. Scope and limitations
5. Background and literature survey
6. Gap and proposed solution
7. System architecture
8. DFD, use case and ER diagrams
9. Technology stack
10. Modules
11. Implementation details
12. Screenshots and live demo
13. Testing and validation
14. Results and discussion
15. Benefits and applications
16. Conclusion
17. Future enhancements
18. References

Introduction: Role of IT in Modern Supply Chains

- Supply chains compete on coordination between independent partners — suppliers, carriers, warehouses.
- IT provides:
  - **Visibility** — one shared, real-time picture
  - **Synchronisation** — partners work to one plan
  - **Automation** — rules replace phone calls
  - **Measurement** — KPIs for continuous improvement
- ERP, WMS, TMS, YMS . . . but the loading dock — the hand-over between transport and warehouse — is often still managed by phone and whiteboard.



Problem Statement

Without dock appointments:

- Trucks arrive at the same peak hours → yard queues, idle docks at other times
- Driver detention and unplanned dock labour
- Double-booking and wrong doors (reefer, hazmat)
- No record of who is late or never arrives
- No data: utilisation? busiest hours? turnaround?

Problem

Build a web platform where warehouses publish dock capacity and carriers self-book **compatible, conflict-free** appointments, while yard events are captured to measure and improve **utilisation, punctuality and turnaround** — securely, for many warehouses and carriers.

Objectives

1. Multi-tenant platform with per-warehouse carrier approval
2. Configurable scheduling: hours, 15–90 min slots, buffers, holidays, maintenance
3. Match loads to docks: inbound/outbound, trailer type, hazmat
4. Guarantee slot capacity under concurrent requests
5. Recurring bookings, rescheduling, automatic waitlist
6. Digitise yard execution; detect late arrivals and no-shows
7. Carrier reliability score and operational KPIs
8. Audit trail and notifications (in-app, e-mail, SMS)
9. Cloud deployment, automated tests, simulation study

Scope and Limitations

Scope

- Users: warehouse managers, carriers, super admin
- Full appointment lifecycle, waitlist, yard events, KPIs, audit, notifications, exports
- One cloud service + managed database, any browser
- Demo data: 2 DCs, 5 typed docks, 6 carriers, 5 weeks

Limitations

- No ERP/WMS/TMS integration (PO/ASN entered manually)
- Gate events recorded manually (no ANPR / IoT)
- Fixed slot length per site
- Super-admin role has no UI yet
- Evaluation on generated / simulated data
- Free hosting: cold start, small CPU

Background: Docks, Yards and Dock KPIs

- **Dock door**: inbound (receiving) or outbound (shipping); capacity per period
- **Equipment**: dry van, reefer (cold chain), flatbed, container; hazmat needs certified bays
- **Yard**: where trucks wait before / after docking
- **Appointment**: a reserved door for a time window

Key performance indicators

- Dock utilisation = booked ÷ offered truck-slots
- On-time arrival rate
- No-show rate
- Turnaround = yard wait + dock time

Literature Survey and Existing Systems

- Port **truck appointment systems** reduce congestion when enforced (Giuliano & O'Brien 2007; Huynh & Walton 2008; Zehendner & Feillet 2014; Phan & Kim 2015)
- Receiving/shipping and **cross-dock scheduling** drive warehouse throughput (Gu et al. 2007; Boysen & Fliedner 2010)
- **Write skew** can double-book under concurrency (Kleppmann 2017)

	Phone / spreadsheet	Port TAS	Dock-scheduling SaaS	WMS/YMS suite	**SmartDock**
Carrier self-service	No	Yes	Yes	Partial	**Yes**
Capacity enforced	Manual	Yes	Yes	Yes	**Atomic, multi-server**
Dock-type / hazmat	Manual	Limited	Varies	Yes	**Yes**
Waitlist / recurring	No	Varies	Varies	Varies	**Yes**
Scorecard & KPIs	No	Terminal KPIs	Yes	Yes	**Yes**
Cost	Effort	Port-run	Subscription	High	**Free, open source**

Gap in Existing Systems and Proposed Solution

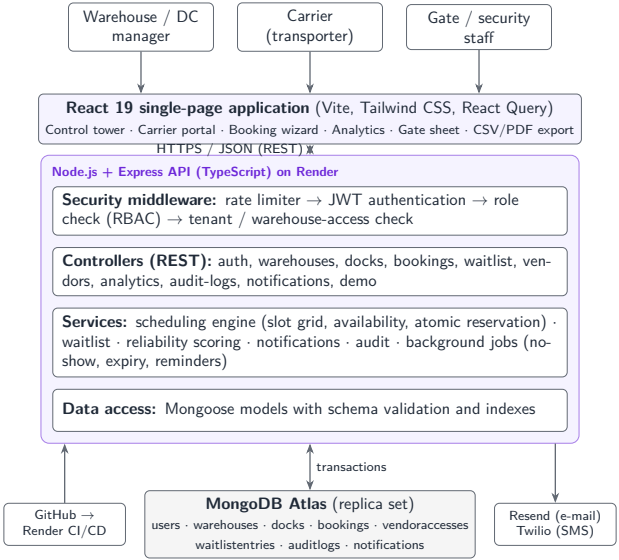
Gap

- SMEs stuck between manual methods and expensive enterprise suites
- Port systems are not built for multi-tenant warehouses with mixed docks
- Simple calendars ignore compatibility, capacity and concurrency
- Booking separated from execution → no KPIs, no accountability

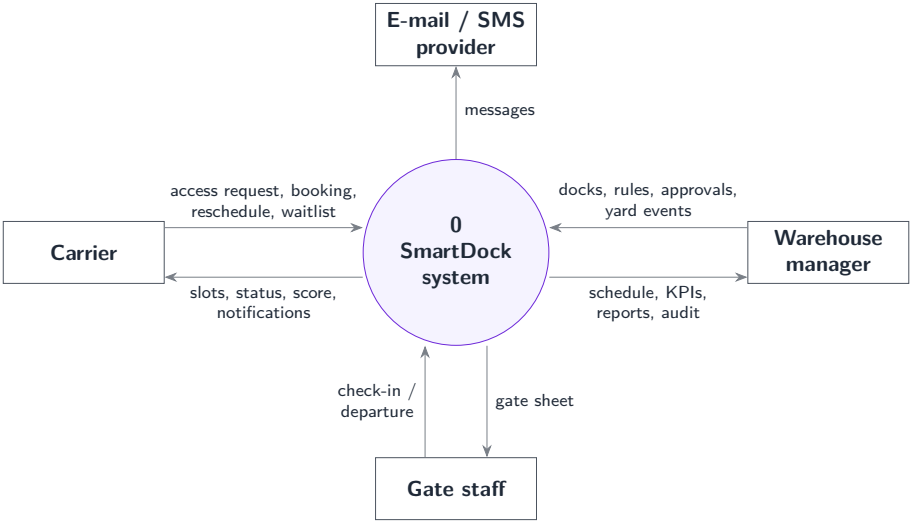
SmartDock

- Per-warehouse rules and carrier access
- Only compatible docks and valid slots; concurrency-safe capacity
- Full lifecycle: request → approve → gate → dock → depart; reschedule, series, waitlist
- KPIs, reliability score, reports, audit, notifications
- Free, open source, one-file cloud deployment

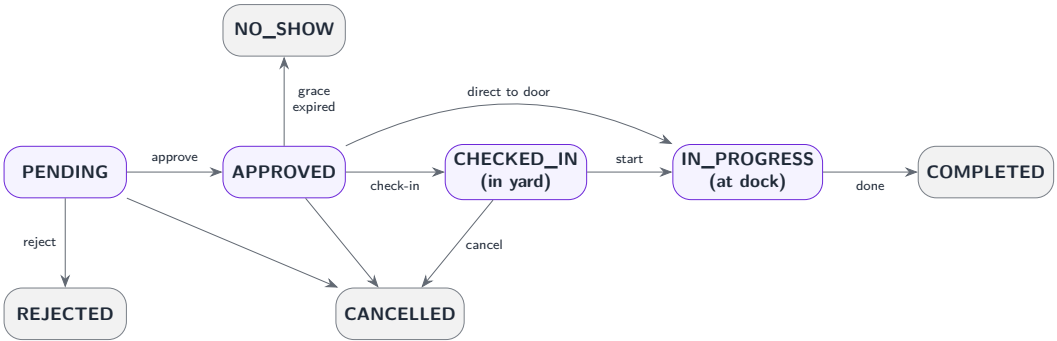
System Architecture



Context Diagram (DFD Level 0)

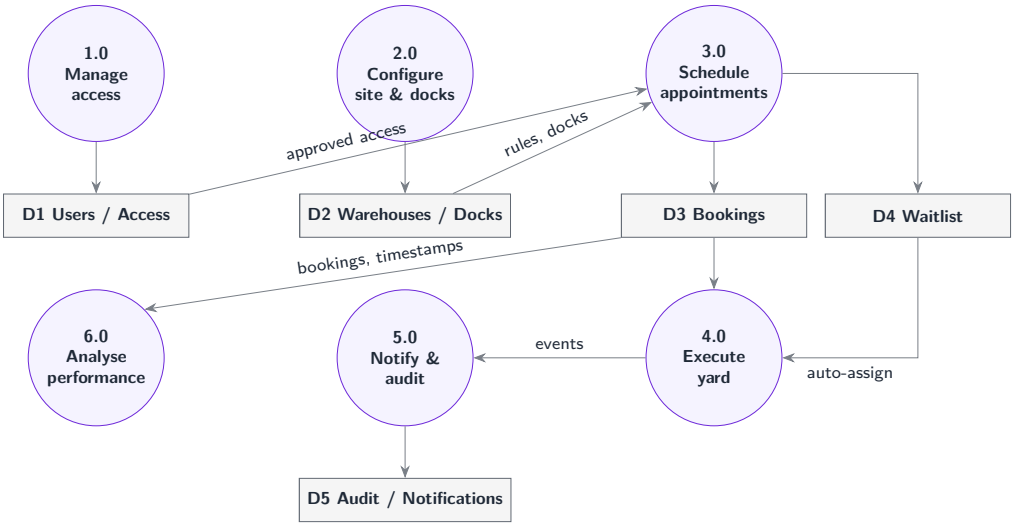


Appointment Lifecycle (State Machine)

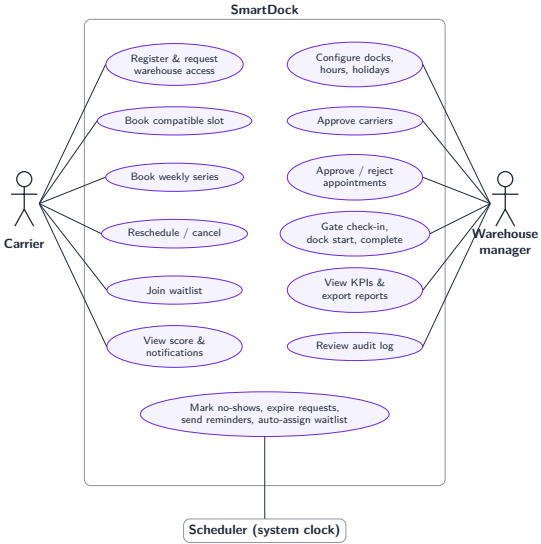


COMPLETED, CANCELLED, REJECTED and NO_SHOW are terminal: released capacity can never be silently re-occupied. Late / on-time is recorded at gate check-in; NO_SHOW is set automatically after the grace period.

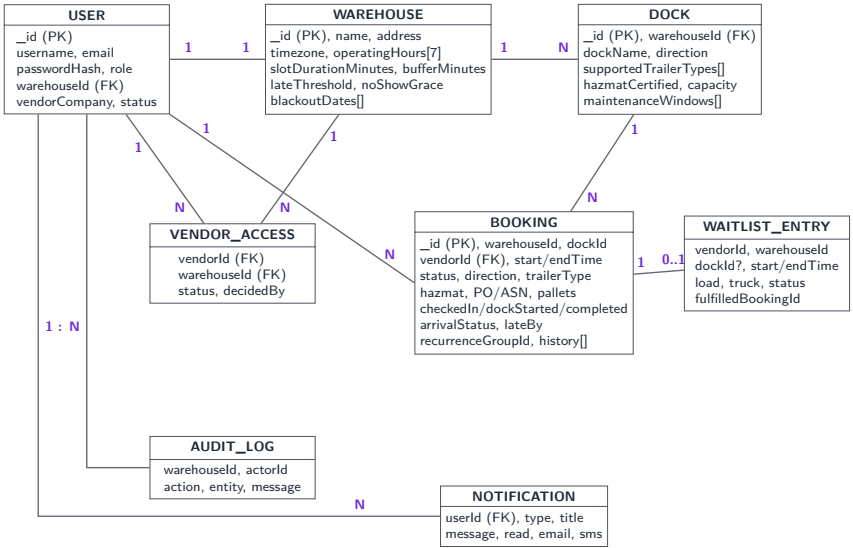
Data Flow Diagram (Level 1)



Use Case Diagram



Entity–Relationship Diagram



Technology Stack

Language	TypeScript (client and server)
Frontend	React 19, Vite, Tailwind CSS, React Query, React Hook Form + Zod, jsPDF
Backend	Node.js 22, Express, Zod validation, JWT, bcrypt, rate limiting, Winston
Database	MongoDB Atlas (replica set), Mongoose — transactions, indexes, aggregations
APIs	REST/JSON (≈40 endpoints); Resend (e-mail), Twilio (SMS)
Cloud & DevOps	Render (Blueprint `render.yaml`, auto-deploy from GitHub), Atlas M0
Testing & tools	In-memory replica set + 2 API instances (58 checks); Git, VS Code, Python, LaTeX

Modules

1. **Identity & access** — roles, per-warehouse approval
2. **Site & docks** — doors, equipment, hours, rules, holidays, maintenance
3. **Scheduling engine** — slot grid, availability, atomic reservation
4. **Booking** — wizard, weekly series, reschedule
5. **Waitlist** — FIFO auto-assignment
6. **Yard execution** — gate check-in, dock start, complete, no-show
7. **Carrier performance** — reliability score, tiers A–D
8. **Analytics** — KPIs, heatmap, CSV/PDF, gate sheet
9. **Notifications** — in-app, e-mail, SMS, reminders
10. **Audit trail** — who did what, when

Implementation: Key Logic

Slot grid (site time zone, DST-safe)

$$s_k = o + k(L + B), \quad e_k = s_k + L \le c$$

L = slot length, B = buffer (30 + 15 ⇒ 08:00, 08:45, . . .)
Compatibility: direction matches (or both) ∧ trailer type supported ∧ (not hazmat ∨ certified)
Arrival: late if check-in − slot start > threshold (15 min)
Background jobs (every minute): no-shows, expiry, 24 h reminders, waitlist expiry

Carrier reliability score

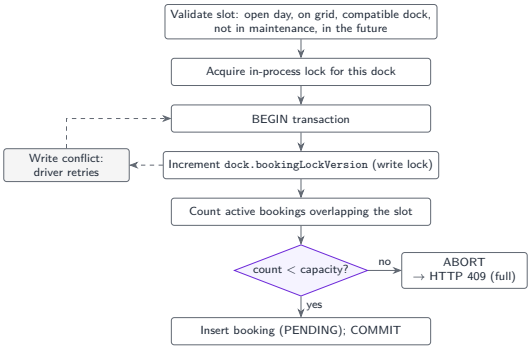
$$R = \frac{\text{on-time} + 0.5 \times \text{late}}{\text{arrivals} + \text{no-shows}} \times 100$$

A ≥ 90 · B ≥ 75 · C ≥ 50 · D < 50

Waitlist

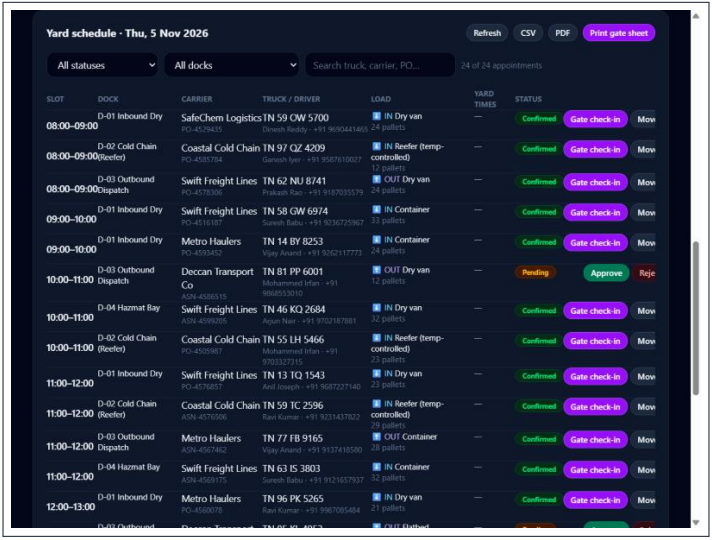
Capacity released (cancel / reject / reschedule) ⇒ oldest compatible waiting carrier booked automatically, inside the same transaction.

Implementation: Concurrency-Safe Capacity Reservation

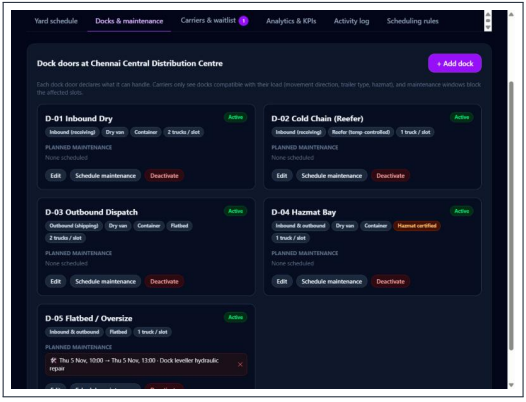


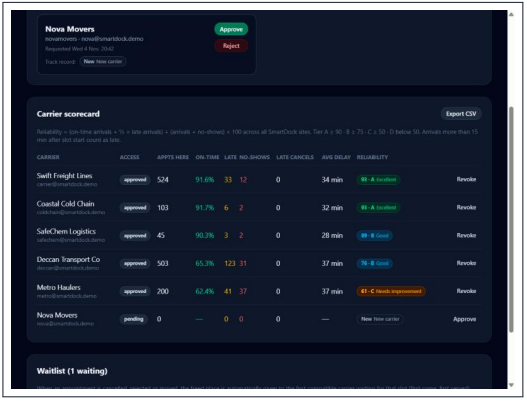
- Naive "count, then insert" lets two requests both see a free place (write skew)
- Even a transaction alone does not help: each reads its own snapshot
- SmartDock: every reservation writes the dock document first ⇒ concurrent transactions conflict, the database retries them, the retry sees the booking
- Plus an in-process lock per dock
- Same routine for bookings, series, reschedules, waitlist

Screens: Yard Schedule (Warehouse Manager)

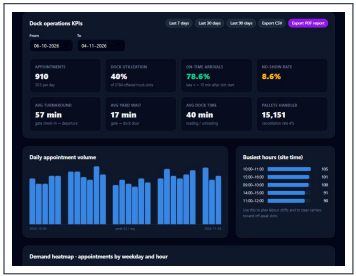


Screens: Docks and Carrier Scorecard

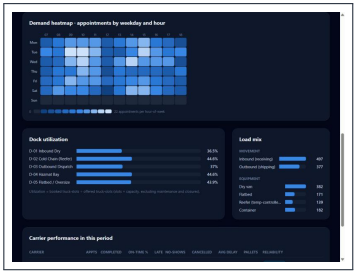




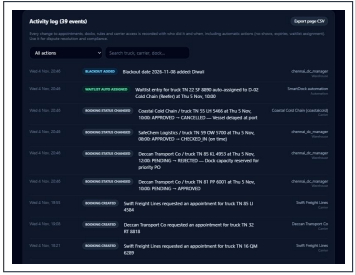
Screens: Analytics, Rules and Audit



KPIs

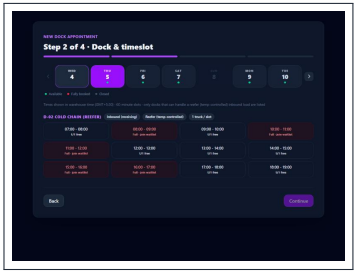


Heatmap & utilisation

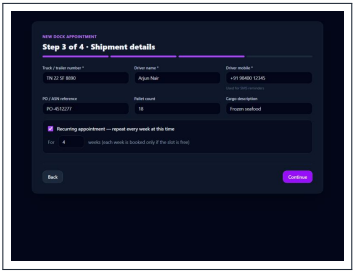


Activity log

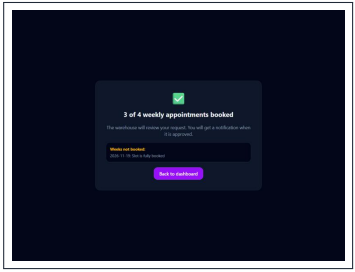
Screens: Carrier Booking Wizard



Compatible docks only; waitlist on full slots



Truck, driver, PO/ASN, weekly series



3 of 4 weeks booked

Live Demo

Links

Live: `https://smartdock.onrender.com`
Code:
`https://github.com/RamaguruS/Dock`

Manager: `admin@smartdock.demo`
Carrier: `carrier@smartdock.demo`
Password: `Demo@123`

Demo flow

1. Carrier books a reefer slot, weekly series; joins a waitlist
2. Manager approves → gate check-in → dock → complete
3. Docks, maintenance, rules, holidays
4. Analytics, PDF report, gate sheet, audit log
5. New carrier (Nova Movers) approval

Testing and Validation

- 58 automated end-to-end checks on an in-memory **replica set** with **two** API servers — all pass
- Security & tenancy, capacity & concurrency, slot rules, dock matching, maintenance, holidays, lifecycle, series, waitlist, reschedule, reliability, audit, analytics
- Live smoke test on the production deployment

Double-booking experiment

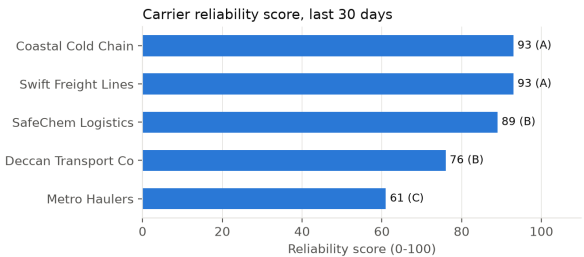
40 simultaneous requests, 2 servers

	Cap. 1	Cap. 3
Naive	15	13
SmartDock	**1**	**3**

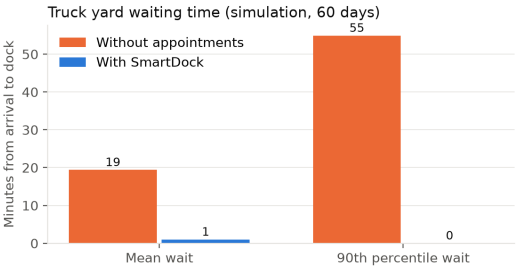
Live API (free tier): slot availability ≈1.3 s, yard schedule ≈1.6 s, 30-day analytics ≈4.3 s (server in USA, database in India)

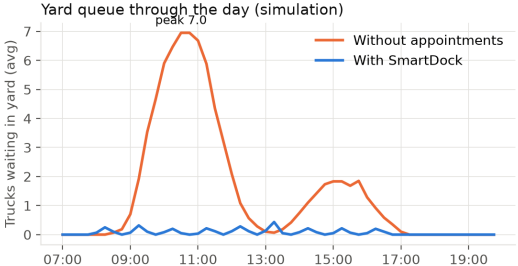
Results: KPIs from the Deployed System (30 days, demo data)

Appointments	910
Dock utilisation	40.0%
On-time arrival rate	78.6%
No-show rate	8.6%
Avg. turnaround	57 min
yard wait / dock time	17 / 40 min
Pallets handled	15,151
Busiest hours	10–11, 15–16



Results: Before vs. After (Simulation, 6 docks, 60 trucks/day, 60 days)





	Without appointments	With SmartDock
Mean yard wait	19.4 min	**1.0 min**
Trucks waiting > 30 min	27.7%	**0.3%**
Peak yard queue	7.0 trucks	**0.4 trucks**

Benefits and Real-World Applications

- **Cost**: less detention and overtime; labour rostered to known load
- **Visibility**: one live schedule for both sides
- **Speed**: self-service booking; waiting $\approx$19 $\rightarrow$ $\approx$1 min (simulated)
- **Traceability**: PO/ASN, timestamps, full audit trail
- **Reliability**: conflict-free, hazmat-safe; carrier accountability
- **Sustainability**: less idling near the site

Applications

Retail & e-commerce DCs · cold-chain / pharma warehouses · chemical depots · 3PL multi-client sites · JIT manufacturing plants · container freight stations

Conclusion

- Designed, built, tested and deployed a multi-tenant dock appointment and yard management system
- All objectives met: configurable scheduling, dock matching, guaranteed capacity, series, rescheduling, waitlist, yard events, reliability scores, KPIs, audit, notifications
- Verified by 58 automated checks and a concurrency experiment; live on the cloud
- A focused IT system at the warehouse–transport interface turns a phone-based process into a measured, collaborative one

Future Enhancements

- **AI/ML**: predict no-shows and delays; smart overbooking; volume forecasting
- **Optimisation**: automatic door assignment; load-based slot length
- **IoT**: ANPR / RFID gate check-in, dock sensors, reefer temperature
- **Integration**: ERP/WMS/TMS (PO, ASN), webhooks, carrier API
- **Mobile**: driver app / PWA with directions and proof of delivery
- **Scale**: WebSockets, analytics caching, Asian region, super-admin console, CI

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Thank you

Questions?

https://smartdock.onrender.com · https://github.com/RamaguruS/Dock

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